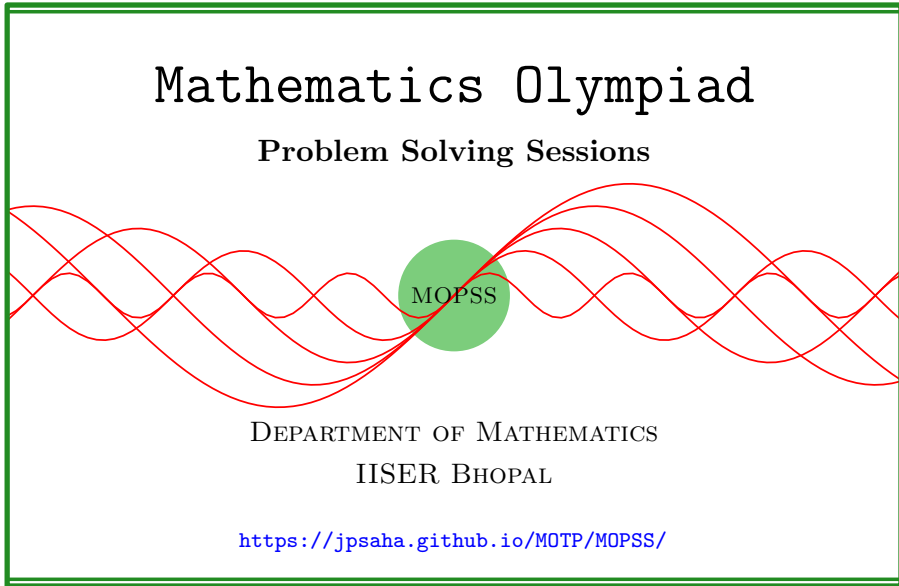


Inclusion-exclusion principle

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

List of problems and examples

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§1 Inclusion-exclusion principle

Example 1.1. How many positive integers not exceeding 1000 are divisible by 7 or 11?

Solution 1. Let A (resp. B) denote the set of integers not exceeding 1000 that are divisible by 7 (resp. 11). Then the size of $A \cup B$ is equal to

$$\#A + \#B - \#A \cap B = \left\lfloor \frac{1000}{7} \right\rfloor + \left\lfloor \frac{1000}{11} \right\rfloor - \left\lfloor \frac{1000}{77} \right\rfloor = 142 + 90 - 12 = 220.$$

■

Example 1.2 (India RMO 1993 P8). I have 6 friends and during a vacation I met them during several dinners. I found that I dined with all the 6 exactly on 1 day; with every 5 of them on 2 days; with every 4 of them on 3 days; with every 3 of them on 4 days; with every 2 of them on 5 days. Further every friend was present at 7 dinners and every friend was absent at 7 dinners. How many dinners did I have alone?

Walkthrough —

- (a) Denote the friends by $F_1, \dots, F_6$. For $1 \leq i \leq 6$, let D_i denote the set of the dinners where F_i was present.
- (b) Show that the total number of dinners is 14.
- (c) Let n denote the number of dinners that I had alone.
- (d) Note that the size of the set $A_1 \cup \dots \cup A_6$ is equal to $14 - n$.

Solution 2. Denote the friends by $F_1, \dots, F_6$. For $1 \leq i \leq 6$, let D_i denote the set of the dinners where F_i was present. Let n denote the number of dinners that I had alone. Since every friend was present in 7 dinners, and was absent in 7 dinners, it follows that the total number of dinners is 14. Consequently, the size of the set $A_1 \cup \dots \cup A_6$ is equal to $14 - n$. By the inclusion-exclusion principle, we obtain

$$14 - n = \sum_{i=1}^6 |A_i| - \sum_{1 \leq i_1 < i_2 \leq 6} |A_{i_1} \cap A_{i_2}|$$

$$\begin{aligned}
& + \sum_{1 \leq i_1 < i_2 < i_3 \leq 6} |A_{i_1} \cap A_{i_2} \cap A_{i_3}| - \sum_{1 \leq i_1 < i_2 < i_3 < i_4 \leq 6} |A_{i_1} \cap A_{i_2} \cap A_{i_3} \cap A_{i_4}| \\
& + \sum_{1 \leq i_1 < i_2 < i_3 < i_4 < i_5 \leq 6} |A_{i_1} \cap A_{i_2} \cap A_{i_3} \cap A_{i_4} \cap A_{i_5}| \\
& - |A_{i_1} \cap A_{i_2} \cap A_{i_3} \cap A_{i_4} \cap A_{i_5} \cap A_{i_6}| \\
& = \binom{6}{1}7 - \binom{6}{2}5 + \binom{6}{3}4 - \binom{6}{4}3 + \binom{6}{5}2 - 1 \\
& = 42 - 75 + 80 - 45 + 12 - 1 \\
& = 13.
\end{aligned}$$

This shows that $n = 1$. In other words, I dined alone only on one day. ■

Example 1.3 (India RMO 2015f P6). From the list of natural numbers $1, 2, 3, \dots$, suppose we remove all multiples of 7, 11 and 13.

- At which position in the resulting list does the number 1002 appear?
- What number occurs in the position 3600?

Solution 3. Let S denote the set of all positive integers none of which is divisible by 7, 11 or 13. Note that the sum of $\#\{x \in S \mid x \leq 1002\}$ and the number of positive integers ≤ 1002 divisible by 7, 11 or 13 is equal to 1002. Moreover, by the inclusion-exclusion principle, the number of positive integers ≤ 1002 divisible by 7, 11 or 13 is equal to

$$\begin{aligned}
& \left\lfloor \frac{1002}{7} \right\rfloor + \left\lfloor \frac{1002}{11} \right\rfloor + \left\lfloor \frac{1002}{13} \right\rfloor - \left\lfloor \frac{1002}{7 \cdot 11} \right\rfloor - \left\lfloor \frac{1002}{11 \cdot 13} \right\rfloor - \left\lfloor \frac{1002}{13 \cdot 7} \right\rfloor + \left\lfloor \frac{1002}{7 \cdot 11 \cdot 13} \right\rfloor \\
& = 143 + 91 + 77 - 13 - 7 - 11 + 1 = 281.
\end{aligned}$$

Note that 1002 belongs to S and hence it appears at the $1002 - 281 = 721$ st position.

Suppose n occurs at the 3600th position. So the number of positive integers $\leq n$ divisible by 7, 11 or 13 is equal to $n - 3600$. Using the inclusion-exclusion principle, we obtain

$$n - 3600 = \left\lfloor \frac{n}{7} \right\rfloor + \left\lfloor \frac{n}{11} \right\rfloor + \left\lfloor \frac{n}{13} \right\rfloor - \left\lfloor \frac{n}{7 \cdot 11} \right\rfloor - \left\lfloor \frac{n}{11 \cdot 13} \right\rfloor - \left\lfloor \frac{n}{13 \cdot 7} \right\rfloor + \left\lfloor \frac{n}{7 \cdot 11 \cdot 13} \right\rfloor. \quad (1)$$

This gives

$$\begin{aligned}
& \frac{n}{7} - 1 + \frac{n}{11} - 1 + \frac{n}{13} - 1 - \frac{n}{7 \cdot 11} - \frac{n}{11 \cdot 13} - \frac{n}{13 \cdot 7} + \frac{n}{7 \cdot 11 \cdot 13} - 1 \\
& \leq n - 3600 \\
& \leq \frac{n}{7} + \frac{n}{11} + \frac{n}{13} - \frac{n}{7 \cdot 11} + 1 - \frac{n}{11 \cdot 13} + 1 - \frac{n}{13 \cdot 7} + 1 + \frac{n}{7 \cdot 11 \cdot 13},
\end{aligned}$$

which is equivalent to

$$\begin{aligned} & -4 \\ & \leq n \left(1 - \frac{1}{7} - \frac{1}{11} - \frac{1}{13} + \frac{1}{7 \cdot 11} + \frac{1}{11 \cdot 13} + \frac{1}{13 \cdot 7} - \frac{1}{7 \cdot 11 \cdot 13} \right) - 3600 \\ & \leq 3. \end{aligned}$$

Noting that

$$\left(1 - \frac{1}{7} - \frac{1}{11} - \frac{1}{13} + \frac{1}{7 \cdot 11} + \frac{1}{11 \cdot 13} + \frac{1}{13 \cdot 7} - \frac{1}{7 \cdot 11 \cdot 13} \right) = \frac{6 \cdot 10 \cdot 12}{7 \cdot 11 \cdot 13},$$

we obtain

$$-4 \times \frac{7 \cdot 11 \cdot 13}{6 \cdot 10 \cdot 12} \leq n - 5 \cdot 7 \cdot 11 \cdot 13 \leq 3 \times \frac{7 \cdot 11 \cdot 13}{6 \cdot 10 \cdot 12},$$

which yields

$$-\frac{7 \cdot 11 \cdot 13}{180} \leq n - 5 \cdot 7 \cdot 11 \cdot 13 \leq \frac{7 \cdot 11 \cdot 13}{240},$$

and consequently, any solution of the Eq. (1) satisfies

$$-5 \leq n - 5 \cdot 7 \cdot 11 \cdot 13 \leq 4.$$

Note that $n = 5 \cdot 7 \cdot 11 \cdot 13$ is the unique solution to

$$n \left(1 - \frac{1}{7} - \frac{1}{11} - \frac{1}{13} + \frac{1}{7 \cdot 11} + \frac{1}{11 \cdot 13} + \frac{1}{13 \cdot 7} - \frac{1}{7 \cdot 11 \cdot 13} \right) - 3600 = 0,$$

and it is a multiple of the pairwise coprime integers 7, 11, 13. Hence, it is a solution to Eq. (1). Also note that $5 \cdot 7 \cdot 11 \cdot 13 - 1$ is also a solution to Eq. (1). Moreover, no integer lying in $[-6 + 5 \cdot 7 \cdot 11 \cdot 13, 4 + 5 \cdot 7 \cdot 11 \cdot 13]$, other than $5 \cdot 7 \cdot 11 \cdot 13 - 1$ and $5 \cdot 7 \cdot 11 \cdot 13$ is a solution to Eq. (1). Observe that $5 \cdot 7 \cdot 11 \cdot 13 - 1$ lies in S .

We conclude that the number that occurs in the 3600th position is $5 \cdot 7 \cdot 11 \cdot 13 - 1 = 5004$. ■

Example 1.4 (India RMO 2019b P5). There is a pack of 27 distinct cards, and each card has three values on it. The first value is a shape from $\{\Delta, \square, \odot\}$; the second value is a letter from $\{A, B, C\}$; and the third value is a number from $\{1, 2, 3\}$. In how many ways can we choose an unordered set of 3 cards from the pack, so that no two of the chosen cards have two matching values. For example we can chose $\{\Delta A1, \Delta B2, \odot C3\}$. But we cannot choose $\{\Delta A1, \square B2, \Delta C1\}$.

Solution 4. Let $\mathcal{A}$ denote the set of ordered tuples (u, v, w) with $u, v, w \in (\mathbb{Z}/3\mathbb{Z})^3$ such that no two among u, v, w have two equal coordinates. For $u \in (\mathbb{Z}/3\mathbb{Z})^3$, let $\mathcal{A}_u$ denote the set of ordered tuples lying in $\mathcal{A}$ with the

first coordinate equal to u . Note that $(u, v, w) \mapsto (u, v, w) - (u, u, u)$ defines a bijection between $\mathcal{A}_u$ and $\mathcal{A}_0$. Hence, the number of ways of choosing an unordered set of 3 cards satisfying the given conditions is $\frac{1}{3!} \cdot 27 \cdot |\mathcal{A}_0|$. Note that

$$\begin{aligned} \mathcal{A}_0 = \{ & (0, v, w) \mid v \notin \langle e_1 \rangle \cup \langle e_2 \rangle \cup \langle e_3 \rangle, w \notin \langle e_1 \rangle \cup \langle e_2 \rangle \cup \langle e_3 \rangle, \\ & w \notin (v + \langle e_1 \rangle) \cup (v + \langle e_2 \rangle) \cup (v + \langle e_3 \rangle) \}, \end{aligned}$$

and hence

$$\begin{aligned} |\mathcal{A}_0| &= \sum_{v \notin \langle e_1 \rangle \cup \langle e_2 \rangle \cup \langle e_3 \rangle} \left| (\langle e_1 \rangle \cup \langle e_2 \rangle \cup \langle e_3 \rangle)^c \cap ((v + \langle e_1 \rangle) \cup (v + \langle e_2 \rangle) \cup (v + \langle e_3 \rangle))^c \right|. \end{aligned}$$

For any subset S of $(\mathbb{Z}/3\mathbb{Z})^3$ and an element $v \in (\mathbb{Z}/3\mathbb{Z})^3$, we have

$$\begin{aligned} |S^c \cap (v + S)^c| &= 27 - |(S^c \cap (v + S)^c)^c| \\ &= 27 - |S \cup (v + S)| \\ &= 27 - |S| - |v + S| + |S \cap (v + S)| \\ &= 27 - 7 - 7 + |S \cap (v + S)| \\ &= 13 + |S \cap (v + S)|. \end{aligned}$$

Henceforth, S denotes the subset $\langle e_1 \rangle \cup \langle e_2 \rangle \cup \langle e_3 \rangle$ of $(\mathbb{Z}/3\mathbb{Z})^3$, and v denotes an element of $(\mathbb{Z}/3\mathbb{Z})^3$ lying outside S .

First, let us consider the case when v has exactly two nonzero coordinates. It follows that v lies in exactly one of $\langle e_1 \rangle - \langle e_2 \rangle$, $\langle e_2 \rangle - \langle e_3 \rangle$, $\langle e_3 \rangle - \langle e_1 \rangle$. Observe that the set $S \cap (v + S)$ is equal to the union of $\langle e_i \rangle \cap (v + \langle e_j \rangle)$ for $1 \leq i, j \leq n$. It follows that $S \cap (v + S)$ has size two.

Now, let us consider the case when all coordinates of v are nonzero. Note that $\langle e_i \rangle \cap (v + \langle e_j \rangle)$ is empty for any $1 \leq i, j \leq n$. Hence, so is the set $S \cap (v + S)$.

Note that the number of elements of $(\mathbb{Z}/3\mathbb{Z})^3$ having exactly two nonzero coordinates is $3 \cdot 2 \cdot 2 = 12$, and the number of elements of $(\mathbb{Z}/3\mathbb{Z})^3$ having all coordinates nonzero is $2^3 = 8$. It follows that

$$|\mathcal{A}_0| = 12 \cdot (13 + 2) + 8 \cdot (13 + 0) = 284.$$

Hence, the number of ways of choosing an unordered set of 3 cards satisfying the given conditions is $\frac{1}{3!} \cdot 27 \cdot 284 = 1278$. ■

References

- [Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)